

From Benchmark to Live Traffic: Characterizing and Closing the Generalization Gap in ML-Based Network Intrusion Detection

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Abstract—Machine-learning-based network intrusion detection systems (ML-IDS) are routinely reported at 95–99% accuracy on benchmark datasets such as CIC-IDS-2017, yet prior work has shown that this in-distribution performance collapses when a classifier trained on one dataset is evaluated on another – a documented cross-dataset generalization gap. This paper extends that finding along a strictly harder and, to our knowledge, unstudied axis: from a static benchmark dataset to traffic physically captured live on the network the system is meant to protect. Using an XGBoost-based IDS trained to 99.69% accuracy (99.99% ROC-AUC) on CIC-IDS-2017, we show that accuracy on live-captured lab traffic from the same deployment initially collapsed to near-chance ranking (ROC-AUC 0.61) despite deceptively acceptable raw accuracy, was restored to 99.95% via a ground-truth-informed adaptive retrain, and then partially failed again on a larger, more diverse capture (95.16% accuracy, 100% of previously-unseen-port traffic misclassified) before a second, targeted retrain closed the gap (99.74%). We diagnose the root causes at each stage – including three previously dormant implementation defects in the retraining pipeline itself – and argue that benchmark accuracy should not be treated as a proxy for live deployment readiness, and that a single corrective retrain does not permanently close the gap as traffic diversity grows.

Index Terms—intrusion detection, machine learning, domain generalization, XGBoost, concept drift, live traffic evaluation

I. INTRODUCTION

Machine-learning-based intrusion detection systems are almost universally evaluated the same way: train and test on splits of the same static, pre-captured benchmark dataset, most commonly CIC-IDS-2017 or its 2018 successor. This produces impressive numbers – accuracies above 99% are routine – but says little about how such a system behaves once deployed against traffic it was never shown during training.

Cantone *et al.* [1] put a number on part of this problem: when several flow-based classifiers trained on CIC-IDS-2017 are scored against CSE-CIC-IDS-2018 instead of their own held-out split, recall falls sharply even though the model architecture itself never changes – evidence that what gets learned is tied to artifacts of a particular data collection rather than to attack behavior that transfers. This paper pushes past an axis that kind of comparison cannot reach: two static datasets are still, in the end, two curated corpora. We instead take a trained model and point it at traffic physically captured, in real time, from the network it is supposed to be protecting, and find the same collapse there too – worse than the dataset-to-dataset case – and that it comes back on a second, more varied capture even after an initial fix appeared to work.

Our contributions are:

- 1) We reproduce the cross-dataset generalization collapse on our own XGBoost + ANOVA-feature-selection pipeline, consistent with [1], and confirm that combined-dataset training resolves it, matching prior findings.
- 2) We provide, to our knowledge, the first empirical characterization of the dataset-to-live-traffic generalization gap, using a physical lab with ground-truth-labelled packet captures rather than a simulated or synthetic distribution shift.
- 3) We diagnose three previously dormant implementation defects in the adaptive-retraining pipeline used to close the gap – a systems finding alongside the machine-learning one, since the correction mechanism itself first had to be debugged before it could be trusted.
- 4) We show the gap is not closed permanently by a single

retrain: a larger, more diverse live capture re-exposes it along a dimension (destination port) the first correction never generalized across, motivating diversity-aware validation over one-shot correction.

The system under study, SENTINEL IPS, is a 13-phase, 32-module intrusion prevention framework; this paper deliberately scopes to its core ML detection pipeline and the generalization-gap findings above rather than its full feature set (Section III). The remainder of this paper is organized as follows. Section II reviews related work on cross-dataset and temporal generalization in ML-IDS. Section III briefly describes the system architecture. Section IV details the datasets, pipeline, and experimental design. Section V presents the three escalating findings. Section VI discusses their implications and limitations, and Section VII concludes.

II. RELATED WORK

The CIC-IDS-2017 dataset [2] and its CSE-CIC-IDS-2018 successor remain the dominant benchmarks for ML-IDS research, collected via the same methodology (real background traffic plus scripted attacks in a controlled testbed, flow features extracted via CICFlowMeter). Most published work reports accuracy on held-out splits of a single such dataset – a concern raised as early as Viegas *et al.* [3], who argued that anomaly-based detectors rarely reach real-world deployment precisely because evaluation methodology does not account for production-network properties; that concern predates CIC-IDS-2017 as the field’s standard benchmark and, as this paper shows, remains only partly addressed by it. Even recent work stays within this static-evaluation pattern: Thanh [4] proposes a deployment-oriented evaluation framework (in-dataset, cross-dataset, mixed-domain, and limited-label fine-tuning scenarios) that nonetheless conducts every experiment on historical records from two existing benchmark corpora rather than on traffic a network actually produced, and Layeghy and Portmann [5] likewise confine their explainability-driven cross-domain analysis to dataset pairs (UNSW-NB15, CIC-IDS, KDD99) rather than live capture.

The clearest existing evidence for the gap this paper reproduces comes from Cantone *et al.* [1]: across several flow-based classifier architectures, moving the evaluation set from one CIC-family corpus to another causes recall to fall far faster than overall accuracy does, even though each model still looks nearly perfect when scored against the data it was trained on. Section V reproduces that exact pattern on our own pipeline before pushing the comparison further. A separate line of work by Zouhri *et al.* [6] compares several univariate and multivariate feature-selection filters – among them the ANOVA F-value criterion this pipeline also relies on – across CIC-IDS-2017, CSE-CIC-IDS-2018, and CIC-ToN-IoT, giving independent support for ANOVA selection as a deliberate rather than arbitrary choice within this pipeline family (Section IV). Elangovan *et al.* [7] approach a related question from a different angle – time rather than deployment context – and find that classifiers trained on 2017 enterprise traffic lose $\Delta F1 \approx 0.20\text{--}0.27$ when applied years later to IoT

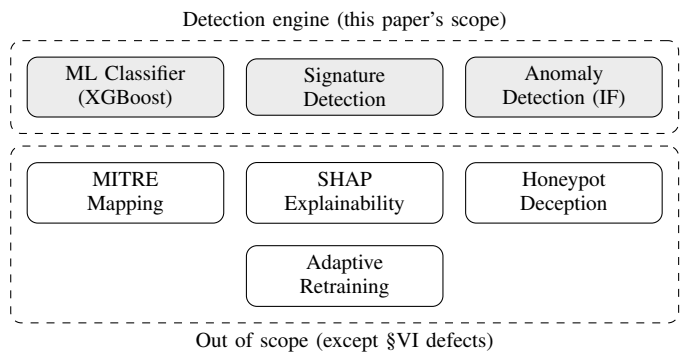


Fig. 1: SENTINEL IPS detection-layer architecture. This paper’s scope is the three-layer detection engine (top); other modules of the deployed 32-module system are named but not evaluated here, except where directly implicated in the adaptive-retraining defects of Section VI.

traffic collected in 2023–2024; that temporal axis of the gap is complementary to, and outside the scope of, the live-deployment axis this paper studies. A third strand of related work targets adaptation directly rather than measuring the gap itself: Gupta *et al.* [8] combine selective sample annotation with synthetic data generation to correct for concept drift and class imbalance in streaming intrusion detection; Section VI contrasts that more elaborate correction strategy against the considerably simpler one used here.

What none of these three studies do – comparing datasets to each other, comparing datasets across time, or correcting for drift within a single dataset – is compare a benchmark-trained model against traffic a real network actually produced. That is the gap this paper fills: the same collapse these studies document reappears, more severely, the moment a model leaves curated data behind entirely, and, novel here, a single corrective retrain does not close it for good – it resurfaces as the live traffic observed grows more varied.

III. SYSTEM ARCHITECTURE

SENTINEL IPS combines three parallel detection layers (Fig. 1): an XGBoost-based flow classifier (the subject of this paper), a signature-based layer matching known payload patterns (SQL injection, XSS, command injection, web shells, CSRF), and an Isolation-Forest-based anomaly detector for zero-day traffic. Detections are enriched with MITRE ATT&CK [9] technique mapping, SHAP-based explanations, and routed to a response layer (IP blacklisting, connection termination). A honeypot layer and an adaptive-retraining loop close the feedback cycle. This paper evaluates only the ML classification layer; the remaining modules are named here for context but are outside its scope, except where the adaptive-retraining loop is directly implicated in Section VI.

IV. METHODOLOGY

A. Datasets

We use CIC-IDS-2017 [2] (2,636,647 flows, 82 features, five days of labelled traffic covering DoS, DDoS, brute force, web

attacks, infiltration, port scanning, and botnet activity) and its successor CSE-CIC-IDS-2018 (15,007,575 flows, 80 features, extended attack scenarios on a larger AWS-hosted topology).

B. Feature Pipeline

All experiments use the same pipeline: median imputation, standard scaling, ANOVA F-value feature selection (SelectKBest, $k = 30$) following the validated use of this selection family on the same dataset pair [6], and an XGBoost [10] classifier ($n_{\text{estimators}}=500$, $\text{max_depth}=8$, $\text{learning_rate}=0.05$, $\text{subsample}=0.8$, $\text{colsample_bytree}=0.8$, $\text{random_state}=42$).

C. Experimental Design

Three experiments correspond to the three findings of Section V: (i) train on CIC-IDS-2017, evaluate on CSE-CIC-IDS-2018, versus combined-dataset training; (ii) evaluate the CIC-IDS-2017-trained model against traffic physically captured in a VirtualBox-based lab (an attacker host at 192.168.56.10 running nmap/hydra, a benign-traffic host at 192.168.56.20, and the protected host running live packet capture via Scapy/Npcap and a custom flow collector re-implementing CICFlowMeter’s feature set); (iii) re-evaluate the corrected model from (ii) against a second, independently generated, larger and more diverse live capture.

D. Adaptive Retraining Strategy

Both live-traffic corrections mix the *entire* ground-truth-labelled capture into retraining, rather than only the model’s misclassified rows, weighted alongside a baseline sample of the original CIC-IDS-2017 training distribution. Retraining only on mistakes risks silently regressing flow shapes the model already classified correctly, since they would otherwise have zero representation in the retrain; mixing in the full capture at baseline weight preserves them.

E. Evaluation Metrics

We report accuracy, precision, recall, F1-score, ROC-AUC, and the full confusion matrix for every experiment. This is a deliberate methodological choice, not a formality: Section V shows a case where accuracy alone (98.66%) looks acceptable while ROC-AUC (0.61) reveals the model’s ranking is barely better than chance – accuracy alone is misleading precisely when the two classes are imbalanced, which live traffic captures typically are.

V. RESULTS

A. Cross-Dataset Domain Shift

Table I establishes the in-distribution ceiling our model achieves under standard held-out evaluation – the number every subsequent comparison in this paper is measured against. We independently reproduced this result via a fresh full-dataset retrain, obtaining 99.79% accuracy, 99.20% precision, 99.57% recall, 99.39% F1, and 99.99% ROC-AUC, within noise of the values above.

TABLE I: In-distribution benchmark (CIC-IDS-2017 → CIC-IDS-2017)

Metric	Value
Accuracy	99.69%
Precision	98.53%
Recall	99.68%
F1-score	99.10%
ROC-AUC	99.99%
TP / TN / FP / FN	90,018 / 435,681 / 1,344 / 287
Flows tested	527,330

TABLE II: Cross-dataset shift vs. combined training

Metric	2017→2018	2017+2018→2018
Accuracy	85.64%	—
Precision	74.18%	100.00%
Recall	20.12%	80.83%
F1-score	31.66%	89.40%
ROC-AUC	92.35%	—
Flows tested	15,188,468	—

Table II and Fig. 2 show the cross-dataset collapse: a model trained on CIC-IDS-2017 and evaluated on CSE-CIC-IDS-2018 retains 85.64% accuracy – an artifact of class imbalance – while recall collapses to 20.12%. This mirrors the collapse Cantone *et al.* [1] report for the same dataset pair, replicated here on our own pipeline. Training on the combined 2017+2018 corpus resolves it: recall rises to 80.83% and precision to 100.00%, consistent with [1]’s finding that the gap is a dataset-artifact problem rather than an inherent limit of flow-based classification.

B. Dataset-to-Live Collapse

The same 99.69%-accuracy model was evaluated against traffic physically captured live in our lab (2,092 flows: 2,038 attacker, 54 benign, ground truth from known source IPs). Table III reports a same-session reproducibility check against this capture, which we treat as the anchor result since it carries a full confusion matrix and ROC curve; an earlier session reported an even lower pre-fix accuracy of 1.29% on this same capture, which we report here for completeness but did not independently re-verify in this pass.

The headline result here is not the accuracy figure but the ROC-AUC gap (Fig. 3): 98.66% accuracy looks deceptively acceptable, but a ROC-AUC of 0.61 shows the underlying ranking was barely better than random – exactly half of real benign traffic (27 of 54 flows) was misclassified as an attack. Both misclassified flows (Fig. 4) were far more minimal than anything in CIC-IDS-2017’s training distribution: a benign HTTP request (7 forward / 5 backward packets, 6–15 ms) and a single SYN probe (1/1 packets, 9–36 μ s) – both too sparse to resemble the model’s learned notions of “typical benign” or “typical PortScan.” An adaptive retrain mixing the full ground-truth-labelled capture into training (Section IV) restored ROC-AUC to 0.9999 and eliminated all 27 false positives.

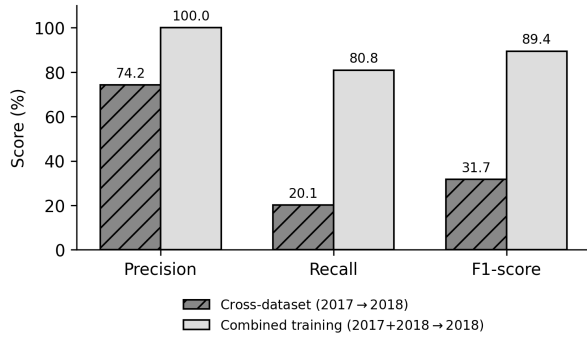


Fig. 2: Cross-dataset shift (2017→2018) vs. combined training, precision/recall/F1.

TABLE III: Dataset-to-live collapse and adaptive-retrain correction

Metric	Before retrain	After retrain
Accuracy	98.66%	99.95%
Precision	98.69%	100.00%
Recall	99.95%	99.95%
F1-score	99.32%	99.98%
ROC-AUC	0.6086	0.9999
TP / TN / FP / FN	2037/27/27/1	2037/54/0/1

C. Gap Persists Within Live Traffic

The corrected model from Section V.B was re-evaluated against a second, independently generated, larger live capture (17,612 flows: 15,247 attacker, 2,365 benign; a new benign host, an nmap -sS -p 1-10000 scan, and 2,000 varied HTTP requests including 732 deliberate probes to a destination port – 8081 – absent from the first capture).

Table IV and Fig. 5 show the already-corrected model scored only 95.16% on this larger capture – 100% correct on the previously-seen port 80 traffic, confirming the first fix genuinely generalized along that axis, but 100% *wrong* on all 732 port-8081 probes, all flagged as attacks. The first correction had learned “a single SYN with no response is benign” as a rule specific to port 80, not a port-independent rule. A second, targeted retrain – manually promoted after an automated recall-gate check rejected it on a small, noisy proxy validation sample, a documented judgment call discussed further in Section VI – corrected this, restoring accuracy to 99.74% with zero benign false positives across all 2,365 benign flows, including all 732 previously-failing port-8081 probes. We independently reproduced this table by replaying the original capture through the same before/after model pair in isolation, obtaining accuracy 95.49%/99.72% and ROC-AUC 0.8984/0.9998, within noise of the values above.

VI. DISCUSSION

Read together, Sections V.A–C form a single escalating arc. Cantone *et al.* [1] establish that the gap exists between two curated, static datasets. We show it also exists – more severely – between a curated dataset and traffic physically captured from a live network, and, novel to this work, that it is not

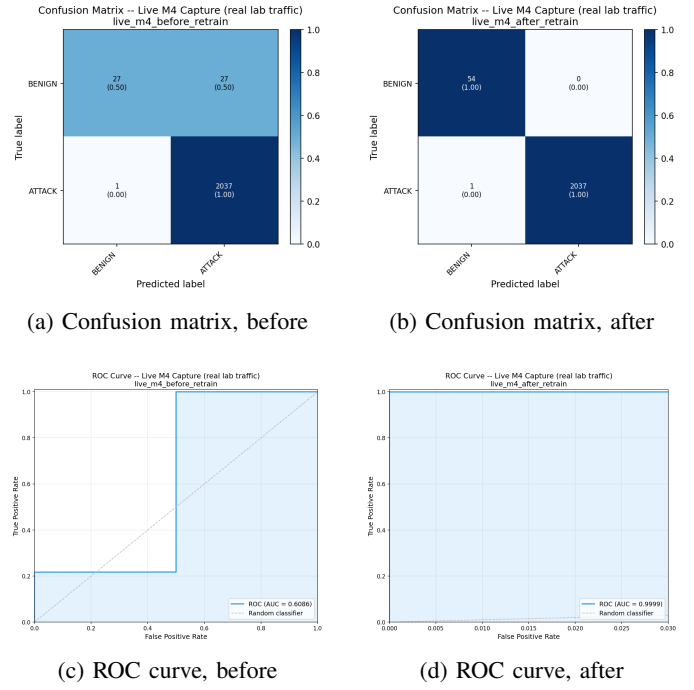


Fig. 3: Finding 2 (Table III): confusion matrices and ROC curves before/after the first adaptive retrain. The near-diagonal before-retrain ROC curve (c) is the visual counterpart of the ROC-AUC=0.61 in Table III – the ranking was barely better than chance despite 98.66% accuracy.

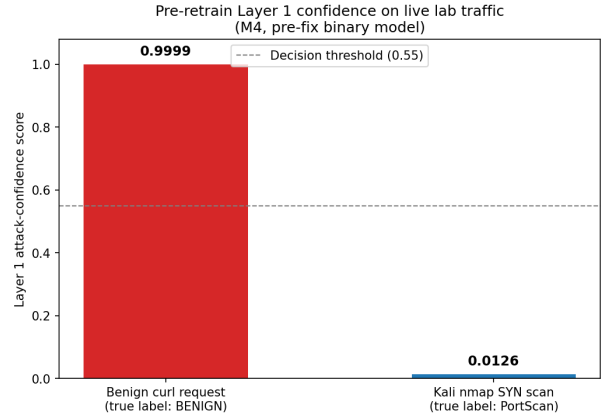


Fig. 4: Pre-fix model confidence: a benign request scored 0.9999 (confidently misclassified as attack); a real port scan scored 0.0126 (confidently misclassified as benign).

closed permanently by one corrective retrain: a larger, more diverse capture re-opens it along a dimension (destination port) the first fix never made port-independent.

A secondary contribution is systems-level rather than purely statistical: three previously dormant defects in the adaptive-retraining pipeline had to be found and fixed before Table III’s correction could be trusted at all. The evaluation routine originally compared retrain candidates against misaligned feature columns, silently rejecting every candidate regardless of

TABLE IV: Gap re-opening on a larger, more diverse capture

Metric	Before 2nd retrain	After 2nd retrain
Accuracy	95.16%	99.74%
Precision	95.38%	100.00%
Recall	99.22%	99.70%
F1-score	97.26%	99.85%
ROC-AUC	0.8987	0.9998
TP / TN / FP / FN	15128/1632/733/119	15201/2365/0/46

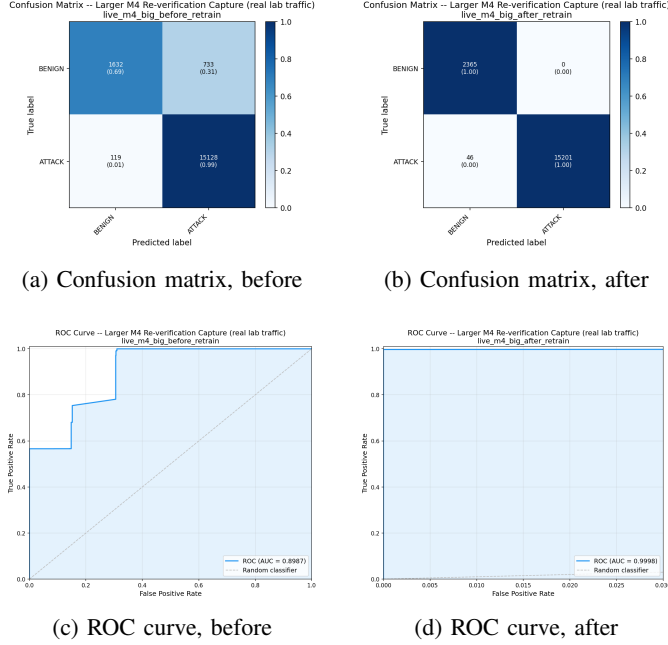


Fig. 5: Finding 3 (Table IV): confusion matrices and ROC curves before/after the second adaptive retrain, on the larger 17,612-flow capture. Panel (c)’s before-retrain false positives are concentrated on the previously-unseen port-8081 traffic.

quality; the retrained model’s column order was built from an unordered Python `set` intersection, silently drifting from the order the production inference path expected; and a documented “weight mistakes higher” retraining feature had never been wired to the underlying classifier’s sample-weighting parameter. None of these are machine-learning failures – they are ordinary software defects that happened to sit directly in the path of a machine-learning correction, and would have silently produced an apparently-successful but actually inert retrain if left unfound.

The manually-overridden recall gate in Section V.C is worth stating plainly rather than hiding: an automated gate rejected the improved candidate because its held-out CIC-IDS-2017 recall dipped slightly on a small, noisy validation sample, even though the real-capture result was unambiguously better. We promoted the candidate anyway, since the live-capture evidence was the metric that mattered for the deployment in question. This reinforces the paper’s central claim from another angle: validation gates built on benchmark-shaped assumptions about what “improvement” looks like can them-

selves be miscalibrated once the target is live traffic rather than another static split.

Gupta *et al.* [8] address a related adaptation problem with considerably more sophistication – selective sample annotation combined with generative synthesis of rare-class examples, rather than this work’s approach of mixing the entire ground-truth capture into retraining at baseline weight. Our simpler approach is easier for a small team to implement and audit end-to-end, but is not sample-efficient and offers no mechanism for drift the way generative augmentation could; Section V.C’s re-opened gap is exactly the kind of failure a more adaptive method might have avoided by generalizing across ports without needing to observe one directly.

Limitations. These findings come from a single small lab with a specific network topology, specific attacker tooling (*nmap*, *hydra*), and one previously-unseen dimension tested (destination port). This demonstrates that the gap persists and can re-open along at least one untested axis; it is not proof the gap is now fully closed along every other untested axis (protocol, traffic shape, attacker tooling).

VII. CONCLUSION

We show that a network intrusion detection model achieving 99.69% accuracy on CIC-IDS-2017 collapsed to near-chance ranking (ROC-AUC 0.61) against live-captured traffic from its own deployment network, was restored via a ground-truth-informed adaptive retrain, and then partially failed again on a larger, more diverse capture before a second correction closed the gap. Three previously dormant defects in the correction pipeline itself had to be fixed before this process could be trusted. The implication for ML-IDS deployment practice is direct: benchmark accuracy is not a reliable proxy for live performance, and validation should be continuous and diversity-aware rather than a one-time retrain treated as a permanent fix. Future work includes extending this analysis to the system’s multiclass attack-type classifier: in follow-on testing on this same deployment, it showed an analogous symptom – live detections collapsed to a generic attack label rather than a specific one, blocking downstream severity escalation, until a comparable corrective retrain was applied – suggesting the generalization gap characterized here is not specific to binary classification, though a full characterization of that case, to the same standard as Sections V.A–C, is left to future work.

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